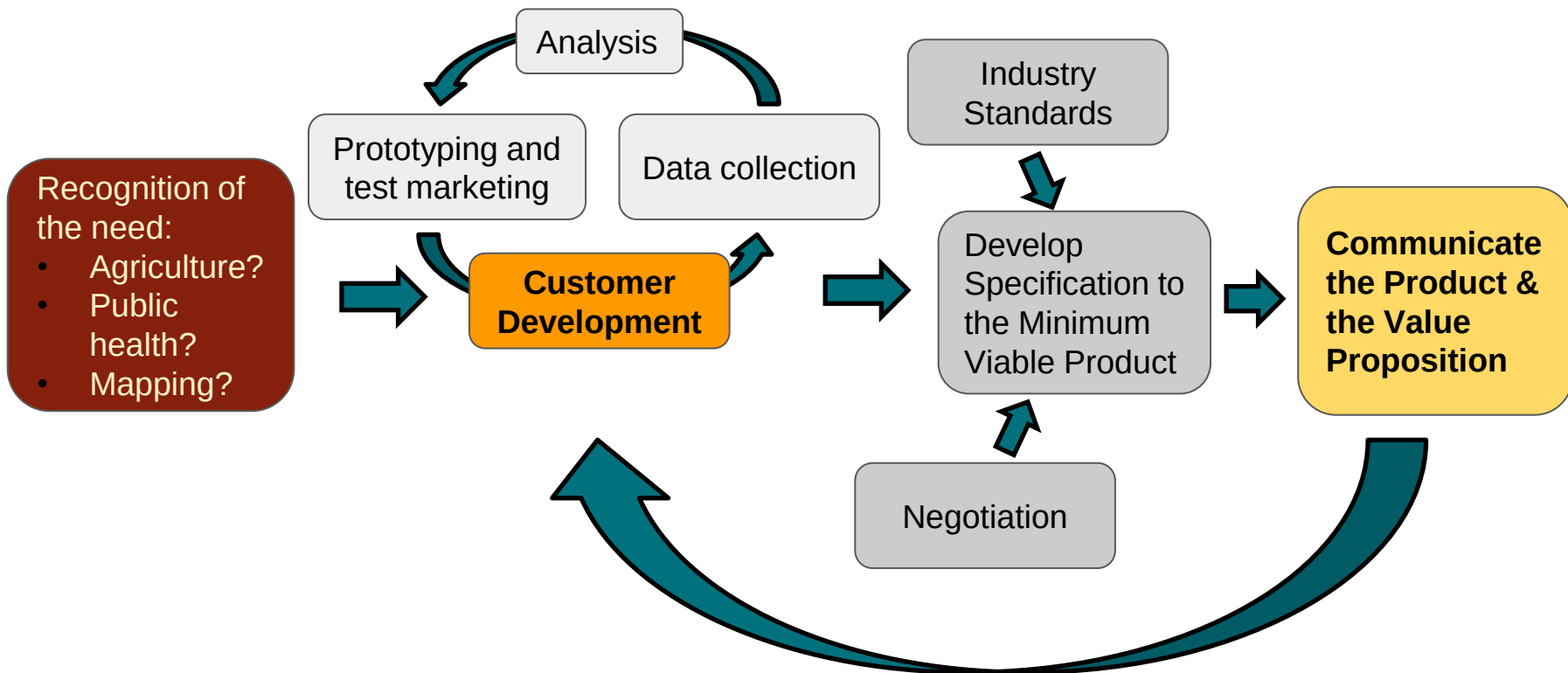




Commercialization, Entrepreneurship and Leadership

Kevin Kochersberger
Virginia Tech

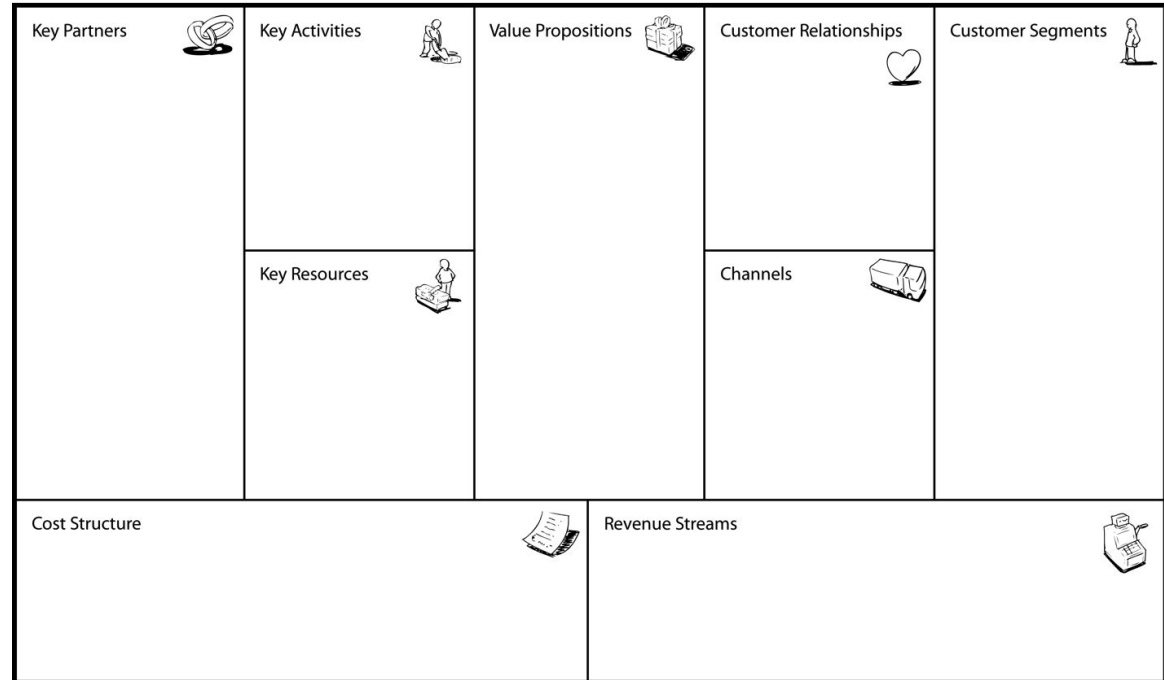
The pipeline for commercialization success



We use two graphical tools to help define the problem we need to solve

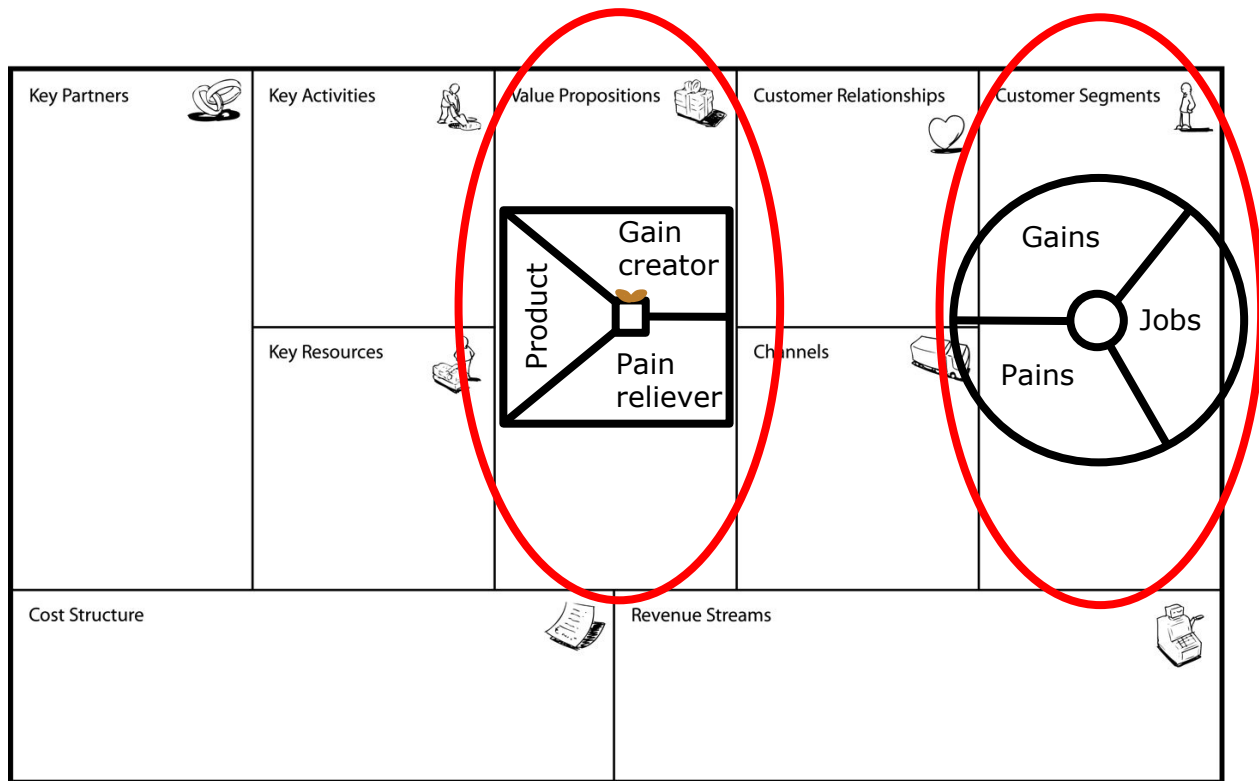
Business Model Canvas

- **The Business Model Canvas (BMC)** is the starting point for most business ideas
- We will work through this template so you understand the components that need to be addressed



We use two graphical tools to help define the problem to be solved

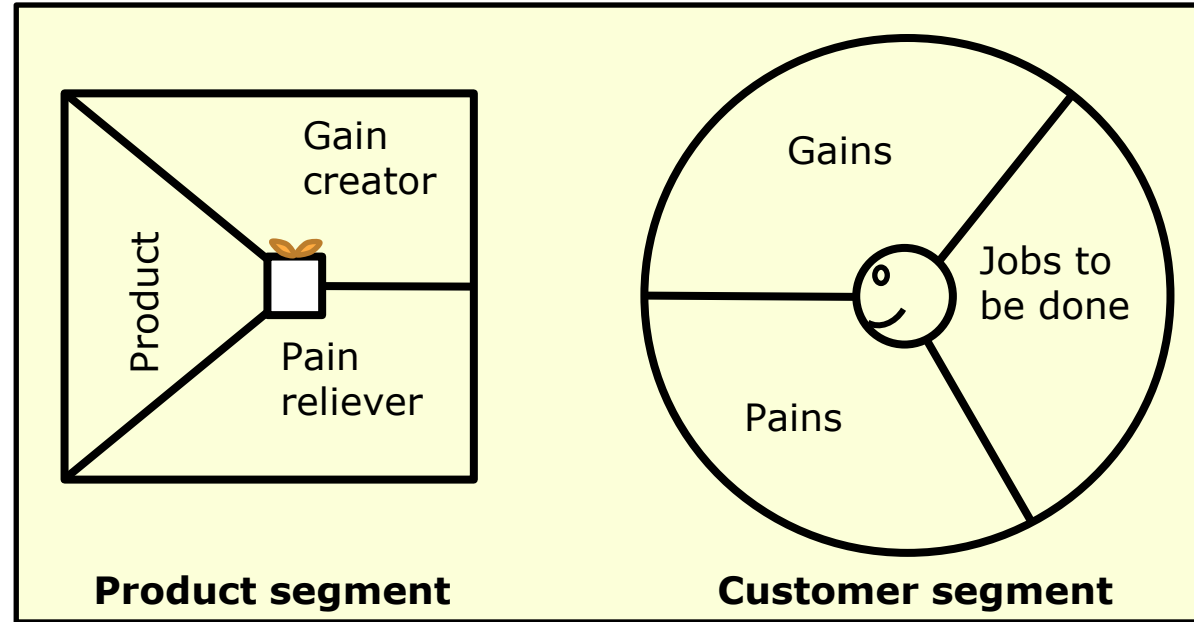
- The **Value Proposition Canvas** is a breakout of two elements in the BMC: the **Value Proposition** and the **Customer Segments**
- Your first task is to **define these two elements** in detail to fully understand what the customer wants



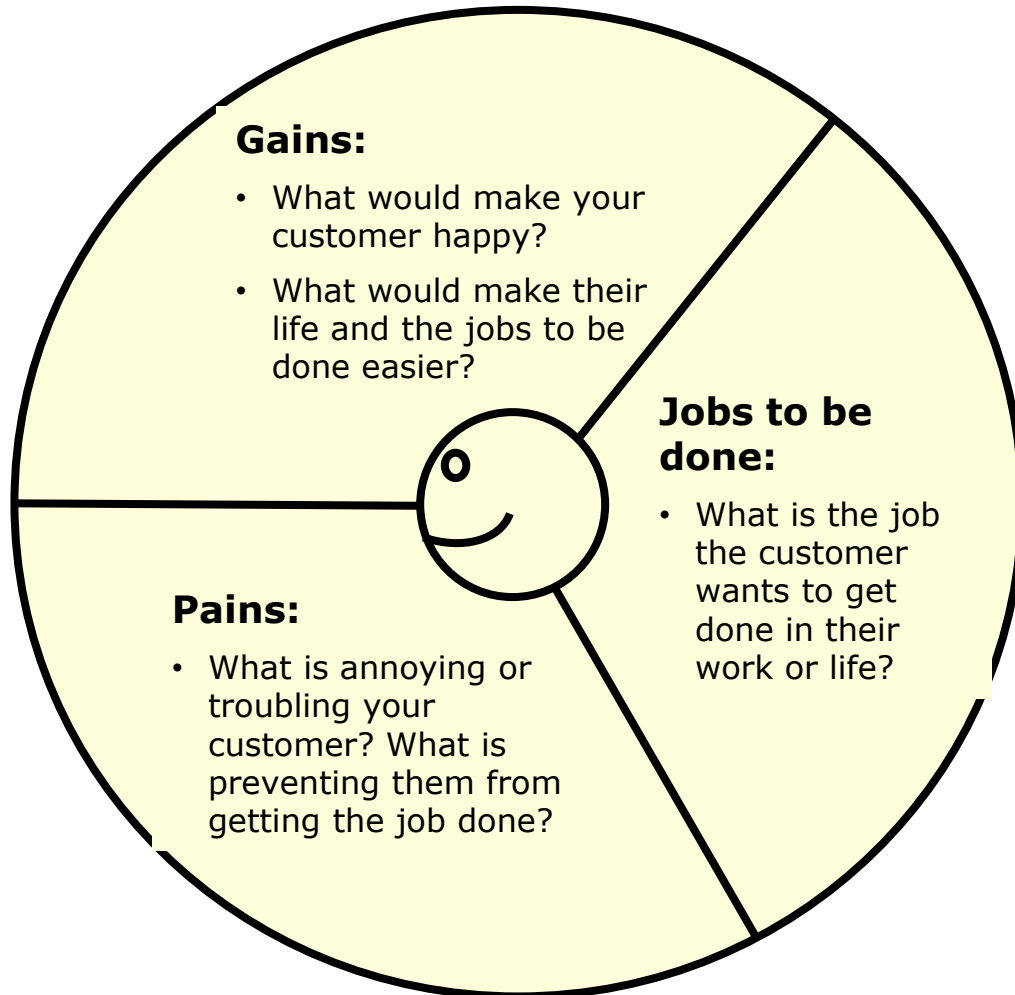
We use two graphical tools to help define the problem to be solved

- The **Value Proposition Canvas** provides a tool to identify the **customer situation and the product fit**
- **Conducting interviews** with potential customers is an important step to identify pains and gains

Value proposition canvas



Value Proposition Canvas The Customer



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Defining Pains & Gains

Gains:

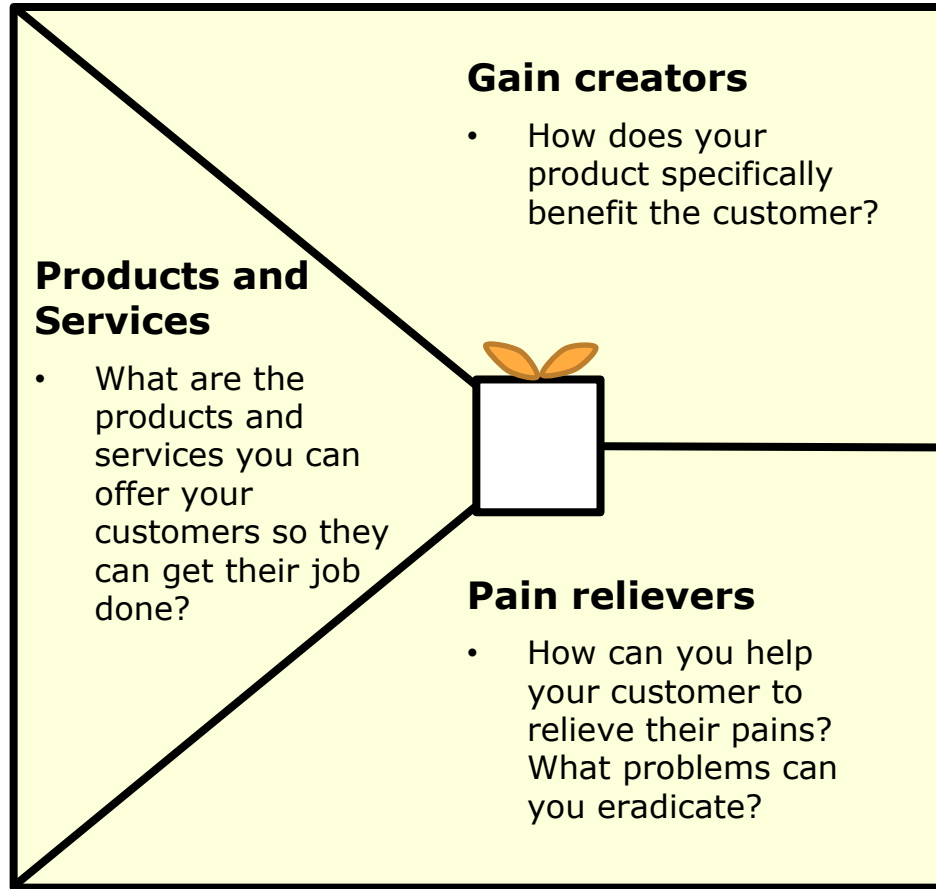
- What would make your customer happy?
- What would make their life and the job to be done easier?

Pains:

- What is annoying or troubling your customer?
- What is preventing them from getting the job done?

Value Proposition Canvas

The Product



Translating Pains & Gains to a viable product

Gain creators:

- What does your product do that specifically benefits the customer?

Pain relievers:

- Where are you reducing a pain experienced by the customer?

The value proposition canvas (VPC) will be populated with responses from your customer interviews

- Always ask open-ended questions
- Don't ask multiple choice questions.
- Ask questions starting with: who, what, why, how
- Not: is, are, would, do you think, should ...
- Ask **why?** then **why?** then **why?** Again

Personas and Problem Scenarios

PERSONA HYPOTHESIS DEVELOPMENT

- Tell me about [yourself in the role of the persona]?
- Tell me about [your area of interest]?
- Tell me your thoughts about [area]?

- What do you see in [area]?
- What do you feel about [area]?
- What do you do in [area]?

(area of interest is the general space that you're looking at)

PROBLEM HYPOTHESIS DEVELOPMENT

- How do you currently [operate in area of interest- if you don't have that yet]? OR Here's what I got on [x]- is that right?
- What's [difficult, annoying] about [area of interest]?
- What are the top [5] hardest things about [area of interest]?
- What are the top 5 things you want to do better this year in [general area of interest]?
- Why is/isn't [your specific area of interest on that list]?

The Art of Customer Interviews

Good

- Play the student card/disarm
- Behavior not feedback
- Insight questions
- Open ended questions
- Listen
- Why-why-why-why-why
- Parrot back to confirm
- Further introductions
- Thank the interviewee
- Record Your Results

Bad

- Selling
- Talking about your technology
- Not listening
- “Learn nothing” questions
- “False positive” questions
- Biased questions
- Interrupting
- Missed opportunities to dig deep

Tips for conducting interviews

- No pitching
 - You should not be perceived as asking for money, or direct marketing a product
- No “feel good” questions
 - The answers to these questions provide no valuable information in selecting a direction for your work
- Pull, don’t push
 - Ask questions that pull information from your customer without specifically addressing the product.
 - Probing and fact-finding questions reveal what is on your customer’s top list of concerns
- Keep your interview pool large
 - You need to talk to many potential customers before you find the average customer concern

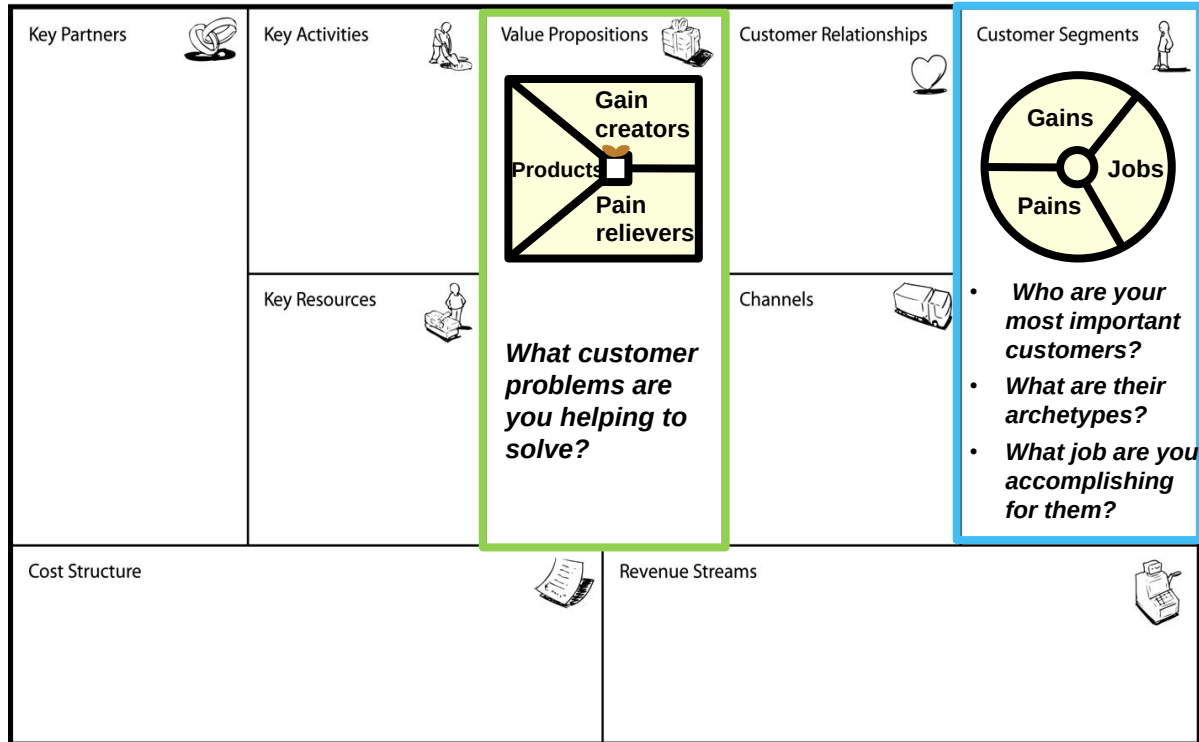


Tips for conducting interviews

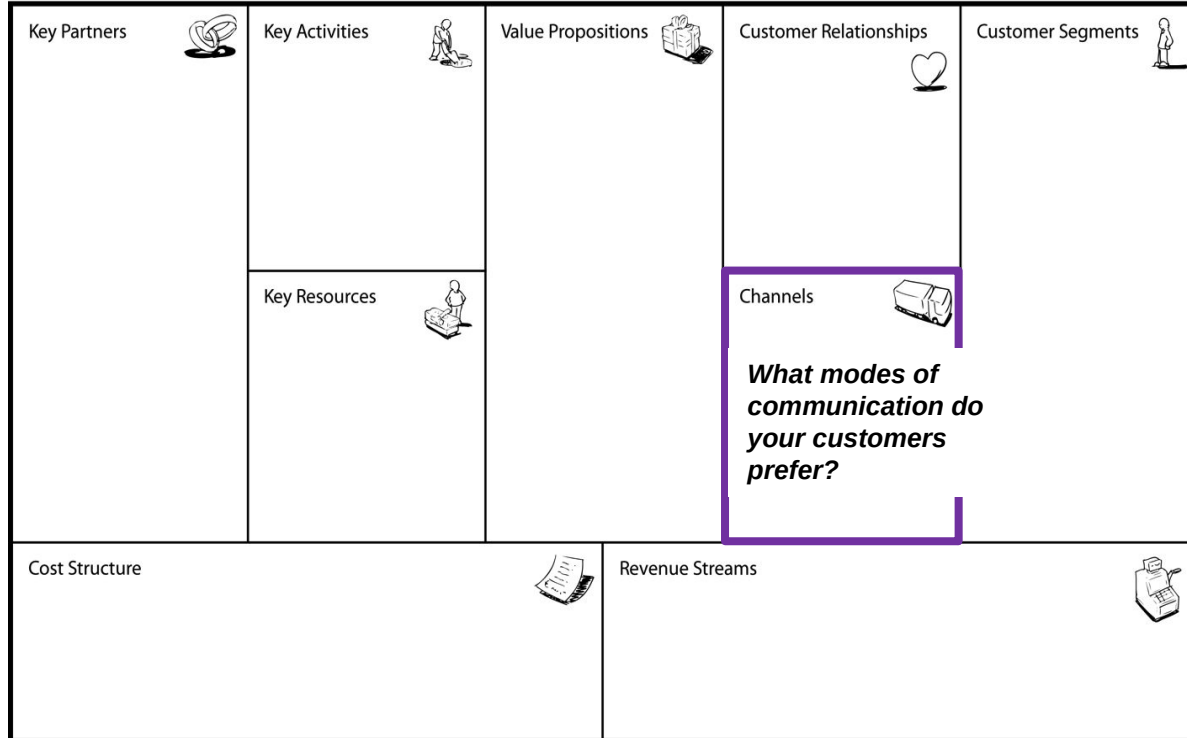
- Past behavior is the best predictor of future behavior
 - Explore historical activity during your customer interview
- Stories are better than statements
 - Establish a dialogue (in-depth conversation) with your interviewee
 - Get to know them, learn about what motivates them, etc.
- Ask “Why?” multiple times to drill down to root problems that need to be solved
 - Every time you ask “why” you force the interviewee to reconsider what bothers them
 - This amounts to a fresh approach to problem inquiry



Once you have defined your value proposition canvas, you should move on to define the business model canvas



Channels are how you access your customers



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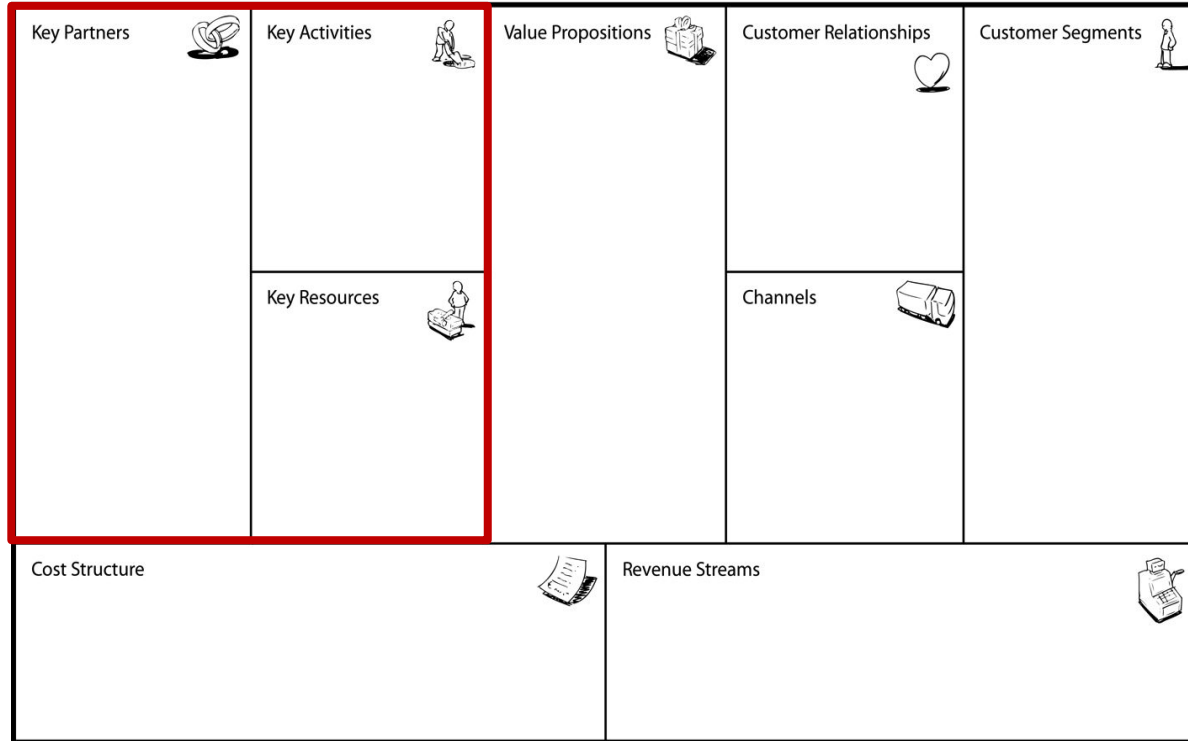
Customer relationships define the short term and long term objectives to find, keep and grow your customers



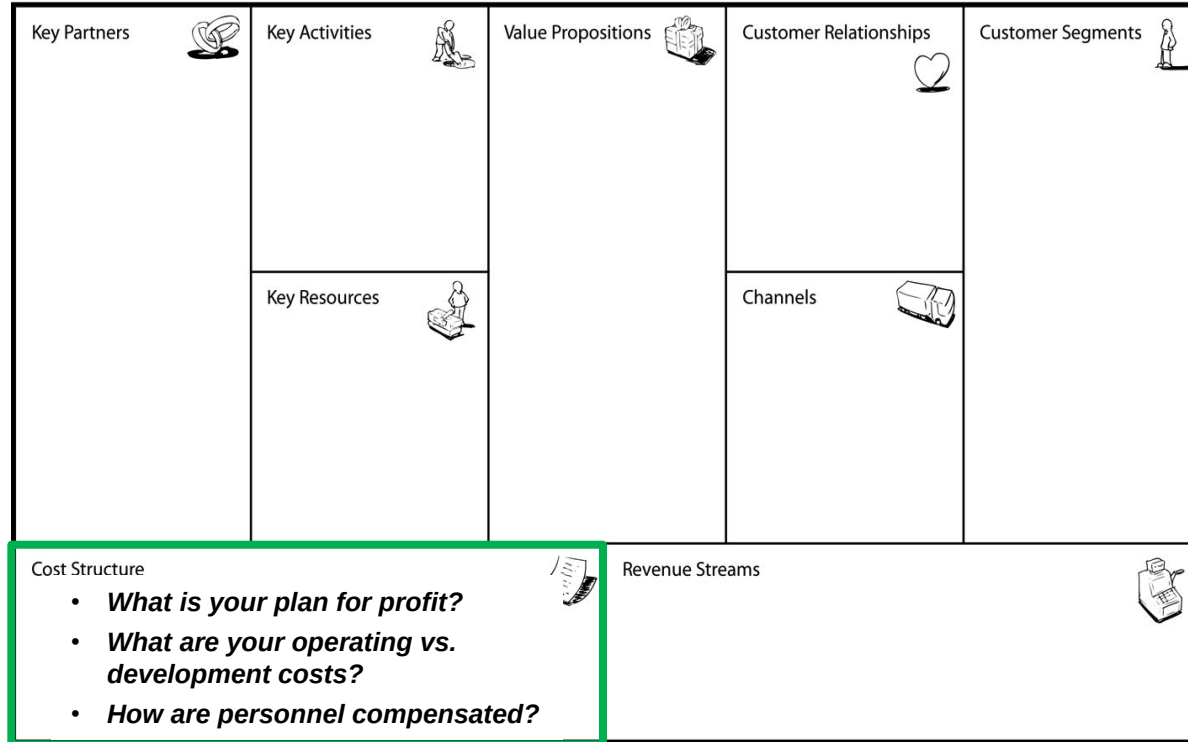
What revenue streams have you identified and will invest in?

Key Partners 	Key Activities 	Value Propositions 	Customer Relationships 	Customer Segments 
	Key Resources 		Channels 	
Cost Structure 		Revenue Streams  • <i>How will you make money?</i> • <i>What is the revenue model?</i> • <i>What are your pricing tactics?</i>		

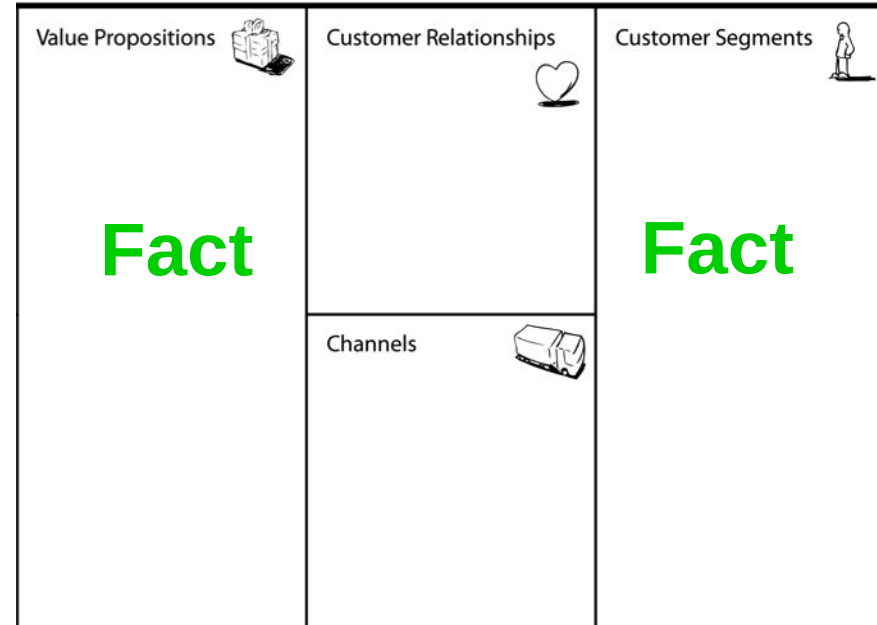
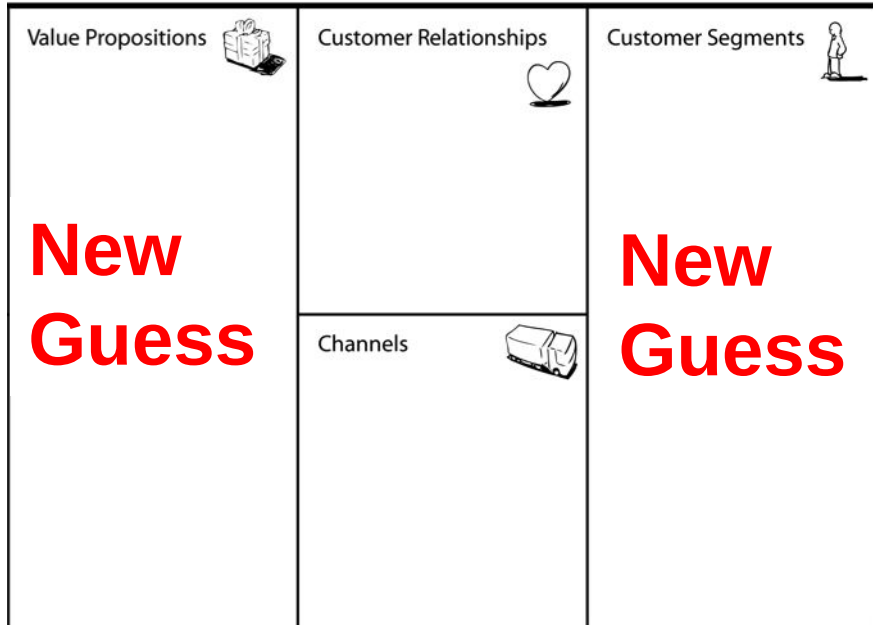
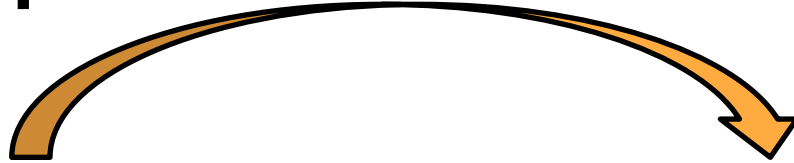
Partners, activities and resources are monitored and updated frequently as company focus evolves



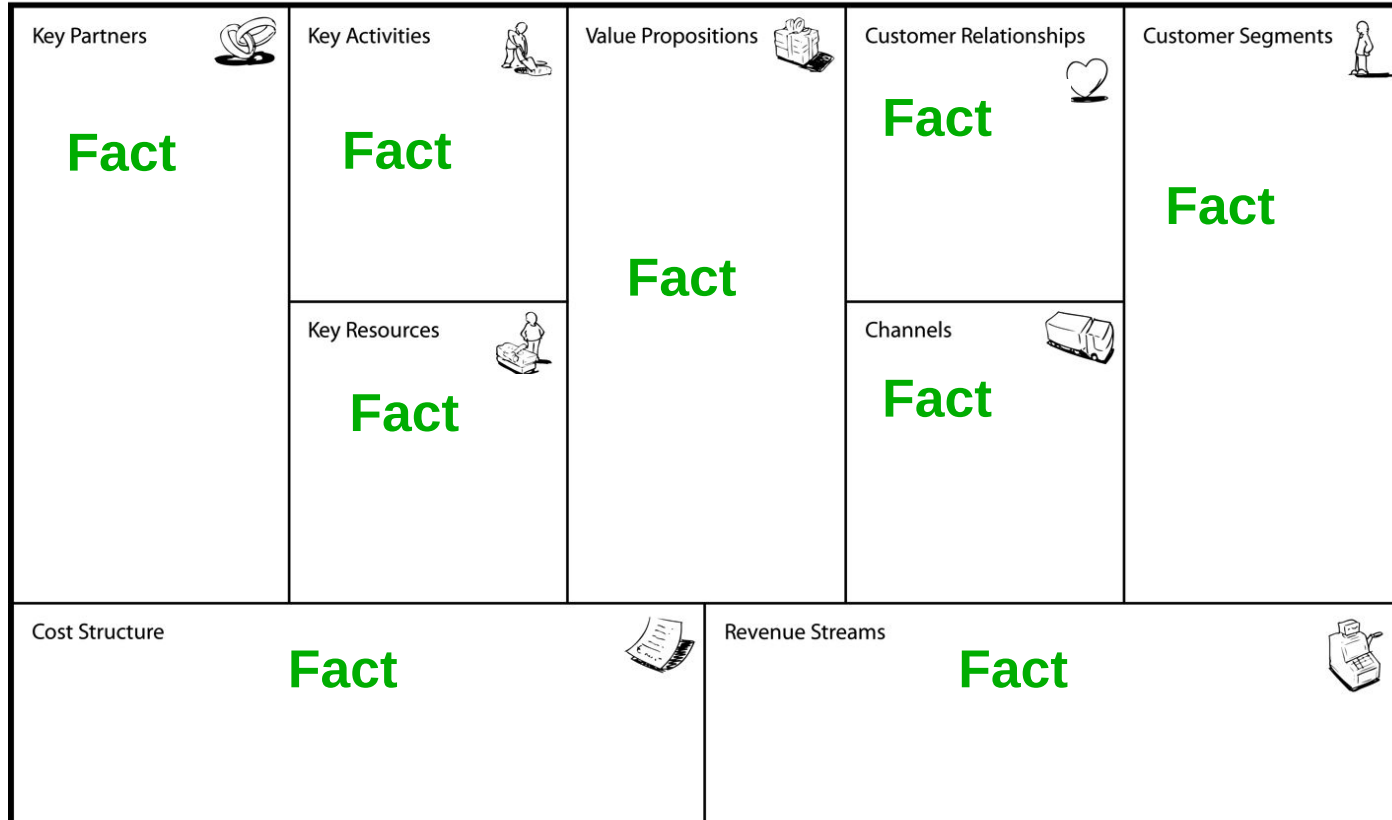
Finally, the cost structure outlines the strategy you will use to effectively manage revenue



With each new set of interview data, the game is to pivot or proceed



The end goal is to fully understand the business canvas





Selling your idea: The Heilmeier Catechism

Kevin Kochersberger
Virginia Tech

Selling your idea: the Heilmeier Catechism

- George Heilmeier from the U.S. Defense Advanced Research Projects Administration (DARPA) proposed this to streamline communication from engineers to management
- In a nutshell, the Catechism is a set of questions designed to give your customer enough information to determine if a proposed project is worth pursuing
- The questions can be used frequently by your project team to self-assess the viability of your idea, and to communicate to investors and other stakeholders about the costs and benefits of the technology

The Heilmeier Catechism

1. What are you trying to do? Articulate your objectives using absolutely no jargon.
2. How is it done today, and what are the limits of current practice?
3. What is new in your approach and why do you think it will be successful?
4. Who cares? If you are successful, what difference will it make?
5. What are the risks?
6. How much will it cost?
7. How long will it take?
8. What are the mid-term and final “exams” to check for success?

Let's test your ability to define and sell a solution using the Heilmeier Catechism

- ***Problem statement: There are no available drones suitable for public safety use***
 - ***Public safety operators frequently repurpose existing drone technology for their specific needs, including:***
 - Imaging and on-site monitoring of situations
 - Carriage of instruments, sensors and other payloads such as spotlight and grappling tools
 - Fly areas affected by natural disaster
 - Delivery of small quantities of medical supplies, water and/or tools to isolated persons
- ***We want to develop a specialized drone that serves the public safety community and meets certain constraints:***
 - Cost
 - Simple to operate
 - Rugged

#1: What are you trying to?

- ***Answer: Develop a drone suitable for public safety use***
 - ***Objectives:***
 - Imaging and on-site monitoring of situations
 - Carriage of instruments, sensors and other payloads such as spotlight and grappling tools
 - Imaging flights in areas affected by natural disaster
 - Deliver small quantities of medical supplies, water and/or tools to isolated persons
 - ***Meet the following constraints:***
 - Cost
 - Simple to operate
 - Rugged



#2: How is it done today, and what are the limits of current practice?

- ***Public safety operators have no choice but to repurpose existing drones for their specific needs***
 - **Selection of payloads is limited** and in many cases the operator must improvise to connect a payload to the aircraft
 - **Specialized sensors such as radiation detectors are not available** for use on drones and this complicates the collection of data
 - **Collection of a large number of images for 3D reconstruction requires a specific flight path** and imaging interval – setting up such a mission is not easy
- ***Compatibility of payloads over many aircraft models does not exist***
 - **Interchangeability** of payloads between aircraft is a priority!



#3: What is new in your approach and why do you think it will be successful?

- *Our aircraft features a universal payload mount that will allow interchangeability of payloads over aircraft platforms*



#3: What is new in your approach and why do you think it will be successful?

- Cost is controlled by using commonly available components and open source software for flight mission planning and data analysis*



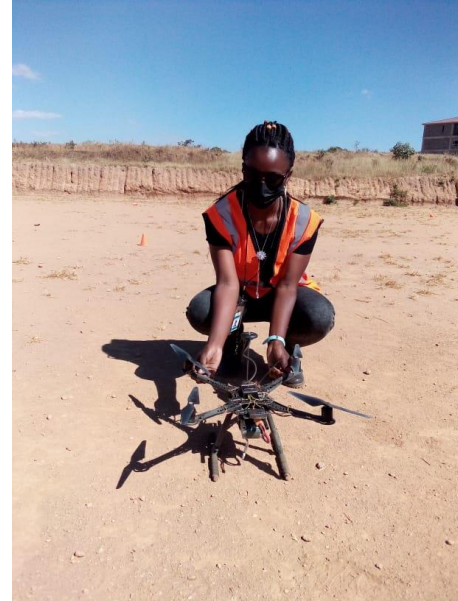
#3: What is new in your approach and why do you think it will be successful?

- ***The aircraft is waterproof***



#4: Who cares? If you are successful, what difference will it make?

- Operators will be able to **save more lives** in a time-limited disaster response
- **Less planning time** for imaging using our streamlined flight planning tools
- **Weather-resistant aircraft** allow flight ops in bad weather conditions
- A public-safety drone operator **training course** will be developed to focus on the public safety mission
- Our project will facilitate **drone adoption** by numerous agencies, including police, fire, public safety, emergency management, and disaster response



#5: What are the risks?

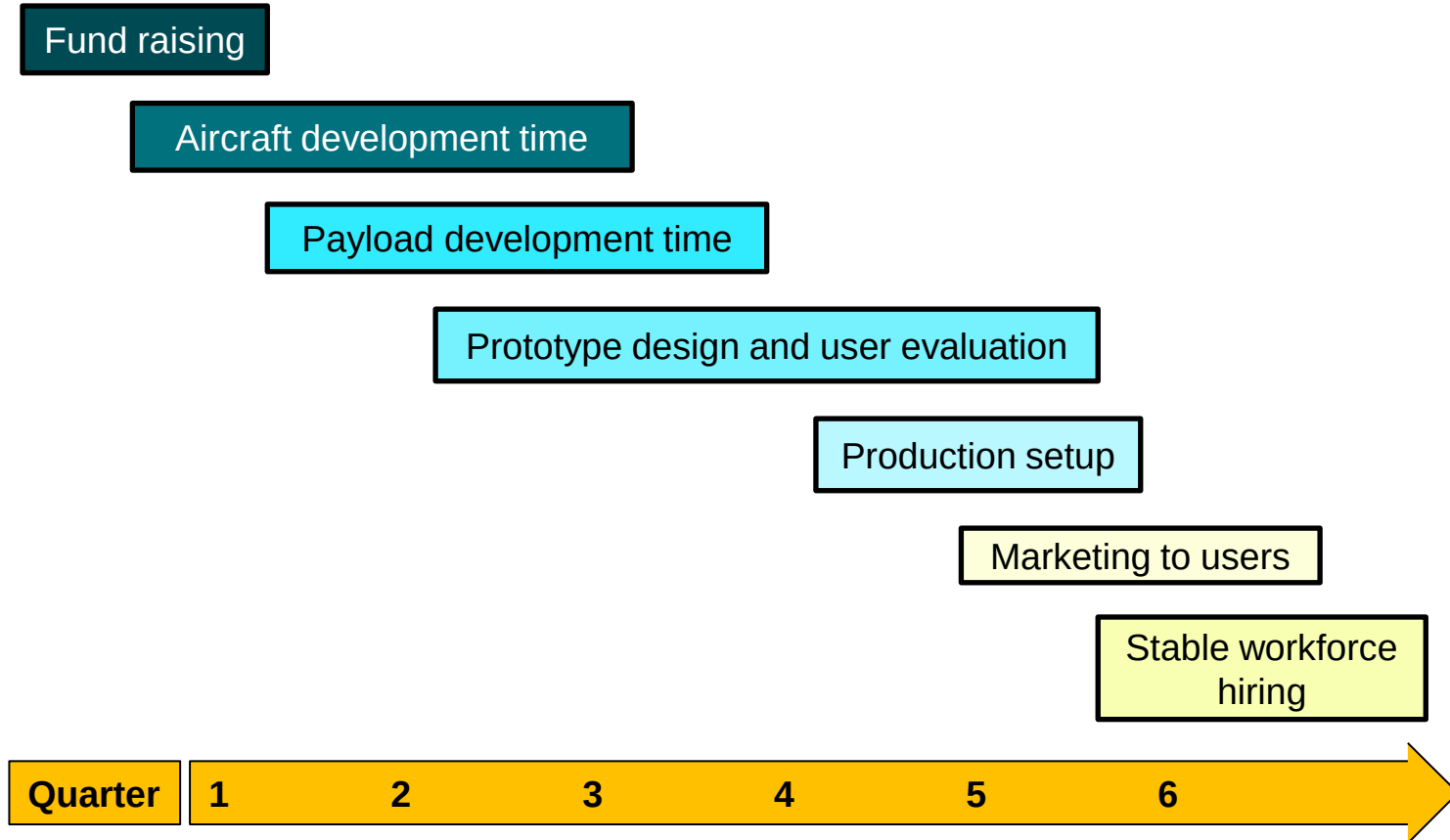
- **Aircraft is not reliable:** need to ensure components are sourced from quality manufacturers
- **Supply chain issues**
- **Affordability**
- **Aircraft requires more development** time than anticipated
- **Payloads require more development** time than anticipated



#6: How much will it cost?

- *Need at least USD 100,000 to start company*

#7: How long will it take?



#8: What are the mid-term and final “exams” to check for success?

- **85% acceptance rate** of the technology by the customer through all stage checks
- **External review committee** consisting of subject matter experts attends quarterly reviews
- At 9 months, we hope to have **1000 commitments to purchase** the aircraft
- At 18 months, **aircraft production is in place**

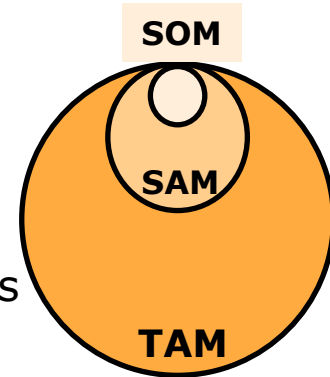


Estimating the market for your product

Kevin Kochersberger
Virginia Tech

Estimating the market for your product

- The **total addressable (or available) market (TAM)** is the market that includes all competitive products
 - It can be expressed as annual expected sales within an observed market
- Methods for estimating the TAM include:
 - **Top-down:** a macro view that considers top-down economics
 - **Bottom-up:** examines data from a limited, local market experience to a broader opportunity
 - **Value theory:** determines the value-added gain from a new product vs. existing
 - Third-party research (risky)



Estimating the market for your product

- The **top-down approach** is generally a good process to follow
 - Start with a population and apply and then logically apply demographic, geographic and economic assumptions to eliminate irrelevant segments
 - This method references well-documented statistical information
- ***If you are selling water, then you would find that the recommended consumption for people is 1.9l of water per day, and there are 7.5 billion people worldwide, and the average cost of water is \$0.32/l. then the TAM would be computed to be \$1.7trillion per year***

Estimating the market for your product

- The **bottom-up approach** involves working on a granular level with specific examples that can be extrapolated to a larger population
 - *Consider that a grocery store chain has a 25% share of the total grocery market, and their revenue is \$100M. The TAM is computed from this data to be \$400M*
- *The problem with this approach is that it can be fairly inaccurate due to assumptions made with a limited dataset*

Estimating the market for your product

- The **value theory approach** is suitable to use when a current market for your product does not exist
 - Consider drone delivery – not much to compare to, so where do you go?
- Customers are asked what they would be willing to pay for a given service
 - Multiplying the purchase price by the number of project users would give an estimate of the market value



The serviceable addressable market (SAM) carves out your company's piece of the pie

- It represents the opportunity that exists within current capabilities and ambitions
- **SAMs** are less than **TAMs** for the following reasons:
 - Geography/culture: ***Food preferences will limit what foods can be sold***
 - Regulation: ***Drone operations are restricted in urban areas***
 - Finite resources: ***Fuel shortages affect supply chains***

The serviceable addressable market (SAM) carves out your company's piece of the pie

- Reports are useful for finding data on a particular region/demographic and customer sector*

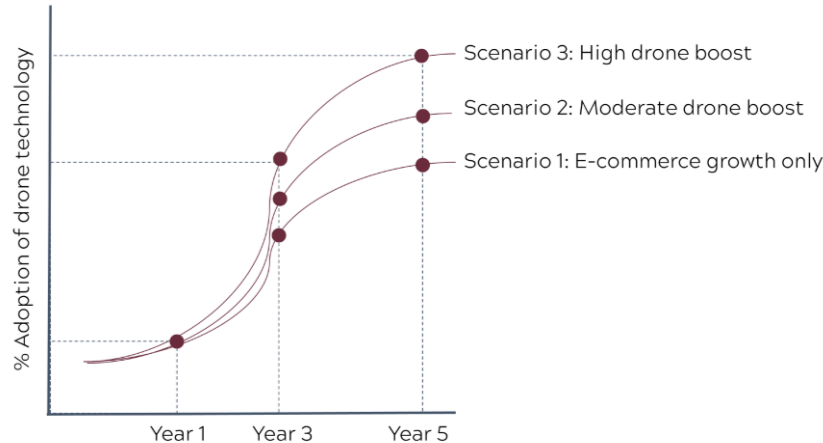
**Lyon-Hill, Measuring the Effects of Drone Delivery in the United States, 2020*

By Year Five, in a single metropolitan area, drone delivery could:

- Serve up to 53.9% of the population
- Recover up to \$582.5 million per year in total time savings for consumers
- Support metropolitan residents who lack access to a vehicle
- Help mobility challenged individuals obtain their prescription medication
- Generate up to \$284,000 per year in new annual sales for a participating local business (up to 250% additional sales compared to a scenario without drones)
- Avoid up to 294 million miles per year in road use and up to 580 car crashes per year
- Reduce up to 113,900 tons per year of CO2 emissions, equivalent to 46,000 acres per year of new forest.

The serviceable addressable market (SAM) carves out your company's piece of the pie

- Reports are useful for finding data on a particular region/demographic and customer sector*
 - Adoption of drone delivery may look similar to adoption of other disruptive technologies

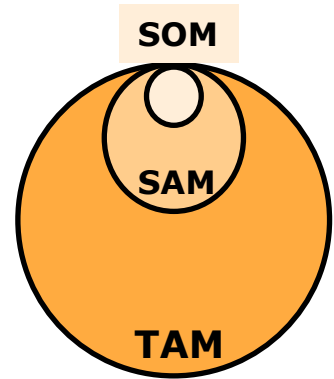


*Lyon-Hill, *Measuring the Effects of Drone Delivery in the United States*, 2020

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Competitive dynamics affect the serviceable obtainable market (SOM)

- This is the market size that the business is realistically targeting to capture within the short term
 - It includes an analysis of the competitive market
- We can look at historic performance and projection forecasts
- Or, choose selected cohorts from the SAM and pick a target market share percentage



Your SOM will be a fraction of the SAM

- The exercise of paring back markets to a realistic subset allows you to better understand the forces that play into product selection

Table 2. Demographic characteristics of the three metropolitan areas¹⁰

Metro area	Population change (2014-2017)	Total population (2017)	Population density (per square mile)	Percent of population ages 18-54	Median household income (2017 inflation adjusted dollars)	Percent of occupied single-family housing units (with yards)
Blacksburg-Christiansburg-Radford, VA	0.2%	181,863	170	57.2%	\$50,409	62%
Austin, TX	9.0%	2,000,590	474	55.7%	\$69,717	61%
Columbus, OH	3.9%	2,023,695	510	51.8%	\$60,170	64%

**Lyon-Hill, Measuring the Effects of Drone Delivery in the United States, 2020
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Project leadership and communication

Kevin Kochersberger
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Project leadership and communication

- Successful teams include the following:
 - Complementary skills
 - Diversity is valued
 - Common purpose and goals
 - Each person understands what others need from her/him
 - Good planning
 - Able to anticipate tasks/deliverables
 - Effective leadership
 - Facilitates activities and allocates resources effectively
 - Appropriate motivators
 - “Chemistry”



Project leadership and communication

- Teams can use a combination of top down or interdisciplinary structure to manage work
 - **Top down management** identifies responsible parties that are accountable for deliverables
 - *There is always a responsible person for deliverables*
 - Interdisciplinary teams rely on team members to manage subtasks
 - There may be team leaders or team facilitators managing smaller elements of a project
- ***Agile teams value individuals and interactions over processes and tools***
 - Customer collaboration over contract negotiation



Teams have basic needs that must be fulfilled

- **Trust is the foundation** to create an effective team
 - Once trust is established, communication becomes more effective
 - Conversely, a violation of trust can result in an immediate breakdown of cooperation
 - Knowing personnel on a social level supports communication



Teams have basic needs that must be fulfilled

- Managing conflict is understood by mature companies as crucial to productivity
 - **Good conflict** arises from passionate concerns of the employees, and the successful channeling of these outcomes to improve a process or condition
 - Managing debate is a tool in good conflict
 - Listening and avoiding defensiveness are recommended
 - Agree to reach consensus
- Conflict resolution can occur in many ways:
 - Confronting / problem solving
 - Compromising
 - Withdrawing
 - Accommodating
 - Collaborating
 - Forcing



Conflict resolution methods

- Conflict resolution via compromising
 - As it states, each party agrees to concessions to reach mutually accepted conclusions
 - When this works it combines all elements of a successful approach
 - When it does not work, a less than full implementation of any solution results
- Conflict resolution via withdrawal or avoidance
 - A concerted effort is made to retract or postpone a given position
 - Useful approach for personality-based or trivial conflicts
 - This does not work when a technical solution is delayed



Conflict resolution methods

- Smoothing or accommodating

- This happens when you are intentionally putting other interests ahead of your interests
 - It can help to resolve a problem by adding time to the solution
 - It may leave a party unsatisfied which can have productivity implications

- Collaborating

- Often considered the best option, it does result in a third solution that may be untested

- Forcing

- An authority figure dictates the path forward
 - Useful when deadlines are imminent and there is little time for debate
 - Usually does nothing to address conflict



Achieving commitment in the team

- Throughout the hierarchy of leadership, we would like to see commitment
 - The leader is committed to the team and project
 - The team is committed to the leader and project
 - Individuals are committed to the team and leadership
- The lack of a leader's commitment to the team will lower expectations throughout the workforce
 - The organization must also support the leadership
- A team's commitment is supported by
 - Trust and conflict management
 - Clear prioritization
 - Structure



Accountability ties the previous elements together

- For an individual, readiness to accept responsibility is accountability
 - There is accountability to the leadership team
 - Rewards and incentives support this
 - Accountability to oneself is also called self-motivation
 - Your goals and objectives drive a focused effort to achieve
 - Accountability to peers
 - Your motivational forces are derived from your relationship with co-workers



The RASCI accountability matrix

- Used to create a mapping of accountability
- R = Responsible (person to complete task)
- A = Accountable (answerable for completion of task)
- S = Support (A subset of R, supporting responsible personnel)
- C = Consulted (Those parties who are specialists or SMEs to a problem)
- I = Informed (To keep everyone up to date with status)



The RASCI accountability matrix

- We can diagram how job duties are mapped to be more effective in managing a project

	Project manager	Project sponsor	Engineers	Test tech
Responsibility X	A	I	R	
Activity 1	A	S		
Responsibility Y	C	R		A
Activity 2	R, A			C
Task alpha	A	R	S	



Focusing on results

- Changing a business focus from internal to external supports a results-focused approach
- Responsibilities that come into focus include:
 - Prioritization: Which tasks are most important to cross the finish line?
 - Clarification: Tasks much gain clarity as they reach final stage development
 - Visibility: The bigger picture must always remain in sight
 - Team members should put team priorities ahead of their own goals
 - Process: Interconnectedness of work is a requirement

